

SS15: Machine learning 2

Content: Background for expert use of graphical models (inference+learning) and development of new algorithms.

VL1. Introduction, motivation, relation to other courses. Graphical models.

Ex1. Introduction to OpenGM inference library.

VL2. Inference in graphical models as integer linear program. Linear programming relaxation.

Ex2. (OpenGM) CPLEX as solver: solver parameters, scalability.

VL3. Convex Optimization. Lagrange duality, complementary slackness.

Ex3. (Blackboard) Lagrange duality for linear and quadratic programs.

VL4. Convex Optimization. First order smooth and non-smooth optimization (gradient, coordinate descent, sub-gradient, Newton method).

Ex4. (CVS with Matlab) Testing different solvers. (Blackboard) Discussion.

VL5. Convex Optimization. Lagrangian (dual) decomposition.

Ex5. (Blackboard) Optimization methods for dual decomposition framework.

VL6. Tree-structured graphical models. Dynamic programming. Reparametrization.

Ex6. (Blackboard) Exercises to dynamic programming.

VL7. Dual of LP relaxation. Sub-gradient and coordinate ascent.

Ex7. (OpenGM) Sub-gradient and coordinate ascent algorithms.

VL8. Dual decomposition for inference with graphical models.

Ex.8 (OpenGM) Tree decomposition - use of different algorithms.

VL9. Sub-modularity. Dual LP as max-flow problem. Sub-modularity as a sufficient condition for non-negativity (solvability).

Ex9. (OpenGM) Graph cut based inference.

VL10. Graph cut based algorithms. Roof duality. Partial optimality.

Ex10. (OpenGM) Graph cut based inference. QPBO.

VL11. Generative learning . Marginal probabilities vs. Gibbs potentials.

Ex11. (Blackboard/Matlab/OpenGM) Generative learning for tree-structured models.

VL12. Discriminative learning. Structural SVM. Cutting plane algorithm.

Ex12. (Matlab/OpenGM) Discriminative learning with different algorithms.

VL13. Structural SVM. Bundle, sub-gradient algorithms. Latent SVM.

Ex13. (Matlab/OpenGM) Discriminative learning with different algorithms.

VL14. Outlook. What we have learned and advanced topics.

EX14. (Matlab/OpenGM) Discriminative learning with different algorithms.